

CSA05-DATABASE MANAGEMENT SYSTEMS

LAB OBSERVATION

EXPERIMENT:17

AIM:

To implement function using program in MySQL.

Procedure and Syntax

Functions:

A function is a subprogram that computes a value.

Creating a function

The CREATE FUNCTION statement is also used in MySQL to support UDFs (user-defined functions). A UDF can be regarded as an external stored function.

MySQL stored function syntax

```
CREATE FUNCTION function_name(param1,param2,...)
```

```
RETURNS datatype
```

```
[NOT] DETERMINISTIC
```

```
.....
```

```
statements
```

```
....
```

Problem 1:

Create a function that returns the level of a customer based on credit limit.

(Use the IF statement to determine the credit limit).

If credit limit > 50000 then customer_level = PLATINUM

If credit limit >= 10000 AND credit limit <= 50000 then customer_level = GOLD

If credit limit < 10000 then customer_level = SILVER

RECURSION in Mysql Procedures

Mysql version should be >= 5.

Have to set system parameters. This means putting the recursion count limit.

```
SET @@GLOBAL.max_sp_recursion_depth = 255;
```

```
SET @@session.max_sp_recursion_depth = 255;
```

Problem 2

Write a recursive MySQL procedure compute the factorial of a number .

SQL QUERIES:

```

mysql> DELIMITER //
mysql> CREATE FUNCTION GetCustomerLevel(p_creditlimit DECIMAL(10,2))
-> RETURNS VARCHAR(20)
-> DETERMINISTIC
-> BEGIN
-> IF p_creditlimit > 50000 THEN
-> RETURN 'PLATINUM';
-> ELSEIF p_creditlimit >= 10000 AND p_creditlimit <= 50000 THEN
-> RETURN 'GOLD';
-> ELSE
-> RETURN 'SILVER';
-> END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
mysql> SELECT GetCustomerLevel(75000) AS customer_level;
+-----+
| customer_level |
+-----+
| PLATINUM      |
+-----+
1 row in set (0.00 sec)

mysql> SET @@GLOBAL.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.00 sec)
mysql> SET @@SESSION.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.00 sec)
mysql> DELIMITER //
mysql> CREATE PROCEDURE Factorial(IN n INT, OUT result BIGINT)
-> BEGIN
-> IF n <= 1 THEN
-> SET result = 1;
-> ELSE
-> CALL Factorial(n - 1, @temp);
-> SET result = n * @temp;
-> END IF;

```

RESULT:

Thus the Functions using program in MySQL executed successfully .